

A Convolution Calculus for Stieltjes Functions

Finite parts, polylogarithm jets, polygamma corrections,
and sharp log-Gamma correlations

Research continuation for Vladimir Reshetnikov's ProveIt project

Prepared with OpenAI assistance

10 October 2026

Abstract

We develop a bilinear extension of the Hurwitz-jet calculus in the ProveIt polylogarithm manuscript. A classical periodic Hurwitz convolution and its oppositely oriented correlation are continued to the pole at order one. Explicit subtraction of the two spatial endpoint singularities gives Hadamard finite-part integrals of arbitrary generalized Stieltjes functions. Both orientations reduce to finite linear combinations of Stieltjes functions at the shift and its complement, with universal coefficients in ordinary zeta values. In particular, for $0 < a < 1$,

$$\begin{aligned}\text{FP} \int_0^1 \psi(x) \psi(\{a - x\}) \, dx &= 2\gamma_1(a) + \zeta(2), \\ \text{FP} \int_0^1 \psi(x) \psi(\{x + a\}) \, dx &= \gamma_1(a) + \gamma_1(1 - a) - 2\zeta(2).\end{aligned}$$

The periodic finite-part distributions form a polynomial convolution algebra generated by the digamma distribution. Bell polynomials describe all its elements and retain the Dirac terms required at coincident shifts. We determine the correction between differentiation and spatial finite parts, obtaining an every-order digamma–polygamma integral. Ordinary log-Gamma correlations provide a second realization of the same calculus; we prove strict convexity, the unique half-period minimum, and an exact convergent expansion with a sharp logarithmic cusp. A complementary moment theorem gives strict Hankel inequalities for the Stieltjes derivative tower. Two precisely scoped manuscript corrections and reproducible symbolic and numerical checks accompany the proofs. Classical ingredients and a rediscovered Dilcher series are explicitly attributed; no claim of historical priority or resolution of the manuscript's remaining Gaussian conjectures is made.

Contents

1	Scope, source audit, and mathematical contribution	3
1.1	Prior results and the limits of the novelty claim	3
2	Conventions and analytic preparation	3
3	The two Hurwitz convolution formulas	4
4	Hadamard finite parts at order one	5
5	Triangular reduction to Stieltjes functions	7
5.1	Low-order identities	8
5.2	A polylogarithm order-derivative identity	8
6	The periodic distribution algebra	9
7	Differentiation of finite parts and polygamma products	11
8	Ordinary log-Gamma correlations and antiderivatives	13
8.1	A critical-order integral	14
9	An exact cusp expansion and strict convexity	14
9.1	Recovery of a known Dilcher series	17
10	All negative-integer jets and Stieltjes antiderivatives	18
10.1	Application to the normalized Stieltjes primitive	20
11	Positive moments in the Stieltjes derivative tower	20
11.1	A sharp asymptotic in derivative order	22
12	Proposed corrections to the current manuscript	23
12.1	The Hurwitz ladder needs an explicit convergence domain	23
12.2	Integer-relation searches do not require an independence theorem	24
13	Further research questions	24
14	Verification, reproducibility, and integration status	26
14.1	Exact algebra and independent numerical checks	26
14.2	A numerical implementation issue found during verification	27
14.3	Delivered sources and proposed integration	27
A	Unequal jets and resonant collision cancellation	27

1 Scope, source audit, and mathematical contribution

The reference snapshot is the ProveIt commit

570b0567f311cf1890865065896be2665f469e4f.

We inspected the current manuscript, its editorial ledger, and the incoming archive register, with detailed attention to the integration, differentiation, conductor, and distribution chapters [1]. The manuscript already proves every-index Stieltjes antiderivative and parameter-derivative formulas, as well as substantial rational-grid and Dirichlet-character reductions. Repeating those identities would add little. The present development studies products at two spatial arguments, their singular limits, and the algebra those limits create.

The main statements proved here are the finite-part coefficient theorem (Theorem 4.2), its two triangular reductions (Theorem 5.2), the convolution algebra (Theorem 6.1), the derivative correction (Theorem 7.1), and the log-Gamma correlation theorem (Theorem 9.1). Each is supplied with an analytic proof. Finite symbolic calculations verify the displayed coefficient tables, while independent real quadrature checks the normalization and signs.

1.1 Prior results and the limits of the novelty claim

Hurwitz’s Fourier expansion, Lerch’s gamma identity, the Hurwitz parameter derivative, and the local polylogarithm expansion are classical [2, 3]. Espinosa and Moll evaluated unshifted and reflected Hurwitz products and used them for log-Gamma integrals [5, Theorem 3.1 and Section 6]. Sun’s normalized periodic Hurwitz convolution is stated in the primary preprint [6, Remark 3.2]; the published formula is also restated in [7, Theorem 1.6]. We give a short Fourier proof and the complex parameter domain needed here. These convolution identities are therefore building blocks, not new discoveries.

The additions relative to the audited manuscript are the explicit separated-singularity finite-part calculus, its every-index triangular reductions, the contact terms and derivative correction, and the resulting sharp shifted-correlation structure. Spectral and distributional treatments of polylogarithms are also established subjects [8]; the proofs below do not establish a first-in-the-literature claim for every specialization. The positive series for γ_1 in Section 9.1 is equivalent to a formula attributed to Dilcher in Brede’s dissertation [9, 10]; it is retained as a consistency check and an application of the correlation geometry.

No audited source states the present convolution problem as an explicit conjecture. Thus this is a proved continuation rather than a claimed resolution of a named existing conjecture. In particular, S_6 and the distinct new frozen S_8 candidate at `s8new:conj:S8` retain their conjectural status. An earlier, different S_8 vector was already rigorously rejected at `research:prop:S8-rejected`; it is not an open conjecture. The already integrated S_4 proof is not reproved here.

2 Conventions and analytic preparation

For $x > 0$ we use

$$\zeta(1+u, x) = \frac{1}{u} + \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \gamma_n(x) u^n, \quad \gamma_0(x) = -\psi(x). \quad (2.1)$$

The prime on $\zeta(s, x)$ denotes differentiation in s ; a superscript $\gamma_n^{(k)}(x)$ or $\psi^{(k)}(x)$ denotes differentiation in x . Set

$$H(u, x) = \zeta(1+u, x) - \frac{1}{u}. \quad (2.2)$$

It is holomorphic in u at zero for fixed $x > 0$. The shift identity gives

$$H(u, x) = x^{-1-u} + H(u, 1+x), \quad \gamma_n(x) = \frac{\log^n x}{x} + \gamma_n(1+x). \quad (2.3)$$

These formulas fix every endpoint subtraction used below.

Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ and write $\{x\}$ for the fractional part. Values at the isolated integers do not affect an ordinary integral; at those points a finite-part or distributional prescription will always be stated. For $0 < a < 1$, put $b = 1 - a$ and define the two orientations

$$C_{f,g}^-(a) = \int_0^1 f(x)g(\{a-x\}) dx, \quad (2.4)$$

$$C_{f,g}^+(a) = \int_0^1 f(x)g(\{x+a\}) dx. \quad (2.5)$$

The minus sign denotes ordinary periodic convolution. The plus sign denotes correlation without complex conjugation. This distinction matters when the two functions differ.

For $k \neq 0$ we take

$$\log(2\pi i k) = \log(2\pi |k|) + \frac{\pi i}{2} \operatorname{sgn} k. \quad (2.6)$$

Polylogarithms on the unit circle away from 1 are their standard analytic continuations from the unit disk. They are entire in the order there. The identities will initially be proved where their Fourier series converge absolutely, then continued with the spatial singularities kept separate.

Lemma 2.1 (Mean and Fourier coefficients). *For $\operatorname{Re} s < 1$, the function $x \mapsto \zeta(s, x)$ is integrable on $(0, 1)$ and has mean zero. For $\operatorname{Re} s < 0$ its absolutely convergent Fourier series is*

$$\zeta(s, x) = \Gamma(1-s) \sum_{k \neq 0} (2\pi i k)^{s-1} e^{2\pi i k x}. \quad (2.7)$$

Proof. The shift identity isolates x^{-s} and a function smooth at zero. Hence integrability follows for $\operatorname{Re} s < 1$. Integrate $\partial_x \zeta(s-1, x) = (1-s)\zeta(s, x)$ and use $x^{1-s} \rightarrow 0$ to obtain zero mean. Equation (2.7) is Hurwitz's Fourier expansion [2, 25.11.9]; its two half-series have terms of magnitude $O(|k|^{\operatorname{Re} s-1})$, proving absolute convergence in the stated half-plane. $\square$

3 The two Hurwitz convolution formulas

Theorem 3.1 (Classical convolution and shifted correlation). *Let $0 < a < 1$, $\operatorname{Re} s < 1$, and $\operatorname{Re} t < 1$. Then*

$$\mathcal{K}_-(s, t; a) := C_{\zeta(s, \cdot), \zeta(t, \cdot)}^-(a) = \frac{\Gamma(1-s)\Gamma(1-t)}{\Gamma(2-s-t)} \zeta(s+t-1, a). \quad (3.1)$$

For $\theta = 2\pi a$ the other orientation is

$$\begin{aligned} \mathcal{K}_+(s, t; a) &= \Gamma(1-s)\Gamma(1-t)(2\pi)^{s+t-2} \\ &\times \left[e^{\pi i(s-t)/2} \operatorname{Li}_{2-s-t}(e^{-i\theta}) + e^{-\pi i(s-t)/2} \operatorname{Li}_{2-s-t}(e^{i\theta}) \right]. \end{aligned} \quad (3.2)$$

Both equalities are jointly holomorphic in this domain. Every fixed finite number of spectral derivatives can be passed through the integrals.

Proof. First suppose $\operatorname{Re} s, \operatorname{Re} t < 0$. For the minus orientation, Fourier orthogonality selects equal frequencies, so the coefficient at $k \neq 0$ is

$$\Gamma(1-s)\Gamma(1-t)(2\pi i k)^{s+t-2}.$$

Comparing with (2.7) at order $s+t-1$ proves (3.1). For the plus orientation, orthogonality selects opposite frequencies, giving

$$\Gamma(1-s)\Gamma(1-t) \sum_{k \neq 0} (2\pi i k)^{s-1} (-2\pi i k)^{t-1} e^{-2\pi i k a}.$$

Separate positive and negative k and use (2.6); the result is exactly (3.2).

In the minus integral the singularities occur at $x = 0^+$ and $x = a^-$. In the plus integral they occur at $x = 0^+$ and $x = b^+$. They are separated for fixed $0 < a < 1$. On a compact subset of $\operatorname{Re} s, \operatorname{Re} t < 1$, local majorants have the form $x^{-\sigma}(1 + |\log x|^r)$ with $\sigma < 1$. They justify joint holomorphy and spectral differentiation under the integral. The right sides are holomorphic in the same domain; for (3.1), $\operatorname{Re}(2-s-t) > 0$ and $s+t-1 \neq 1$. The identity theorem, first in one variable and then the other, extends the formulas throughout this connected product of half-planes. $\square$

The orientation reversal cannot be omitted: at $s = t = 0$ the first formula is $\zeta(-1, a)$, whereas the second is $-\zeta(-1, a)$. At $a = 0$ the plus integral has coincident singularities and requires an additional integrability condition. We keep $0 < a < 1$ in all finite-part formulas.

4 Hadamard finite parts at order one

Definition 4.1 (The two spatial finite parts). For $n, m \geq 0$ and $0 < a < 1$, define

$$\begin{aligned} F_{n,m}^-(a) = \lim_{\epsilon, \eta \downarrow 0} \left\{ \int_{\epsilon}^{a-\eta} \gamma_n(x) \gamma_m(a-x) dx + \int_a^1 \gamma_n(x) \gamma_m(1+a-x) dx \right. \\ \left. + \frac{\gamma_m(a)}{n+1} \log^{n+1} \epsilon + \frac{\gamma_n(a)}{m+1} \log^{m+1} \eta \right\}. \end{aligned} \quad (4.1)$$

With $b = 1 - a$, define

$$\begin{aligned} F_{n,m}^+(a) = \lim_{\epsilon, \eta \downarrow 0} \left\{ \int_{\epsilon}^b \gamma_n(x) \gamma_m(x+a) dx + \int_{b+\eta}^1 \gamma_n(x) \gamma_m(x-b) dx \right. \\ \left. + \frac{\gamma_m(a)}{n+1} \log^{n+1} \epsilon + \frac{\gamma_n(b)}{m+1} \log^{m+1} \eta \right\}. \end{aligned} \quad (4.2)$$

The two cutoffs tend to zero independently. The powers of their logarithms use the actual coordinates displayed here, fixing the normalization.

The limits exist: after the leading singular term is subtracted, each remaining local integrand is $O(1 + |\log x|^{\max(n,m)})$. Thus these definitions can also be evaluated by ordinary convergent integrals.

Set $w = u + v$ and introduce

$$E(u, v) = \frac{\Gamma(1-u)\Gamma(1-v)}{\Gamma(1-w)}. \quad (4.3)$$

The two holomorphic numerator germs are

$$B_-(u, v; a) = -E(u, v)(1 + wH(w, a)), \quad (4.4)$$

and

$$B_+(u, v; a) = \Gamma(1-u)\Gamma(1-v)(2\pi)^w \times \left[e^{\pi i(u-v)/2} \text{Li}_{-w}(e^{-2\pi i a}) + e^{-\pi i(u-v)/2} \text{Li}_{-w}(e^{2\pi i a}) \right]. \quad (4.5)$$

They satisfy $B_{\pm}(0, 0; a) = -1$. Define the mixed divided difference

$$\mathcal{R}_{\pm}(u, v; a) = \frac{B_{\pm}(u, v; a) - B_{\pm}(u, 0; a) - B_{\pm}(0, v; a) + B_{\pm}(0, 0; a)}{uv}. \quad (4.6)$$

It is holomorphic even when $u = 0$ or $v = 0$.

Theorem 4.2 (Every-index finite-part coefficient formula). *For all $n, m \geq 0$ and $0 < a < 1$,*

$$F_{n,m}^{\pm}(a) = (-1)^{n+m} n! m! [u^n v^m] \mathcal{R}_{\pm}(u, v; a) = (-1)^{n+m} n! m! [u^{n+1} v^{m+1}] B_{\pm}(u, v; a). \quad (4.7)$$

Thus (4.5) is an explicit integral identity involving polylogarithm order derivatives at zero, gamma derivatives at one, and arbitrary Stieltjes functions.

Proof. For $\text{Re } u, \text{Re } v < 0$ near zero, the product of the two functions H is integrable. Since the mean of each Hurwitz zeta is zero,

$$C_{H(u, \cdot), H(v, \cdot)}^{\pm}(a) = \mathcal{K}_{\pm}(1+u, 1+v; a) + \frac{1}{uv} = \frac{B_{\pm}(u, v; a) + 1}{uv}. \quad (4.8)$$

The sign of the final $1/(uv)$ follows by expanding the two pole subtractions and using the zero means; it is essential.

Near the first singularity, (2.3) shows that the integrand equals $x^{-1-u}H(v, a)$ plus a locally integrable holomorphic remainder. Integrating this term meromorphically gives residue $-H(v, a)$ at $u = 0$. At the second singularity the residue at $v = 0$ is $-H(u, a)$ in the minus orientation and $-H(u, b)$ in the plus orientation. No other nearby poles occur, since after these terms are subtracted the remainders are bounded by x^{-r} with $r < 1$ on a sufficiently small bidisk.

Consequently the holomorphic regularizations are respectively

$$\frac{B_- + 1}{uv} + \frac{H(v, a)}{u} + \frac{H(u, a)}{v}, \quad \frac{B_+ + 1}{uv} + \frac{H(v, a)}{u} + \frac{H(u, b)}{v}.$$

Equivalently, the boundary identities are

$$\begin{aligned} B_-(u, 0; a) &= -1 - uH(u, a), & B_-(0, v; a) &= -1 - vH(v, a), \\ B_+(u, 0; a) &= -1 - uH(u, b), & B_+(0, v; a) &= -1 - vH(v, a). \end{aligned} \quad (4.9)$$

These yield exactly (4.6).

To identify the coefficients with the specified spatial finite parts, use

$$\int_{\epsilon}^c x^{-1-u} dx = \frac{\epsilon^{-u} - c^{-u}}{u}.$$

The coefficient of $(-u)^n/n!$ in the subtracted endpoint contribution is $\log^{n+1} \epsilon/(n+1)$, with exactly the signs in (4.1)–(4.2). The regular remainders and their spectral derivatives are dominated by integrable logarithmic powers. Coefficient extraction and the two cutoff limits therefore commute. This proves both identities in (4.7). $\square$

5 Triangular reduction to Stieltjes functions

The polylogarithm representation is not the end of the reduction. Introduce

$$T(u, v) = \frac{\Gamma(1-u)\Gamma(1+u+v)}{\Gamma(1+v)}, \quad U(u, v) = T(v, u), \quad V(u, v) = \frac{vT(u, v) + uU(u, v)}{u+v}. \quad (5.1)$$

The quotient defining V is holomorphic: on $v = -u$ both T and U equal 1, so its numerator vanishes. A holomorphic function vanishing on that hyperplane is divisible by $u + v$.

Proposition 5.1 (Gamma-only numerator). *For $b = 1 - a$,*

$$B_+(u, v; a) = -V(u, v) - vT(u, v)H(u+v, a) - uU(u, v)H(u+v, b). \quad (5.2)$$

Proof. Apply the two Hurwitz Fourier equations at arguments a, b and order $1 + w$ to the two polylogarithms in (4.5). Solving this two-by-two linear system first where $\sin \pi w \neq 0$ gives

$$B_+ = -E(u, v) \frac{w}{\sin \pi w} [\sin \pi v \zeta(1+w, a) + \sin \pi u \zeta(1+w, b)].$$

Euler's reflection identity gives

$$E(u, v) \frac{\sin \pi v}{\sin \pi w} = \frac{v}{w} T(u, v), \quad E(u, v) \frac{\sin \pi u}{\sin \pi w} = \frac{u}{w} U(u, v).$$

Substitute $\zeta(1+w, x) = 1/w + H(w, x)$ and cancel. The resulting (5.2) extends across $w = 0$ by holomorphy. $\square$

The relevant gamma quotients have the convergent logarithmic expansions

$$\log E(u, v) = \sum_{r \geq 2} \frac{\zeta(r)}{r} (u^r + v^r - (u+v)^r), \quad (5.3)$$

$$\log T(u, v) = \sum_{r \geq 2} \frac{\zeta(r)}{r} (u^r + (-1)^r ((u+v)^r - v^r)). \quad (5.4)$$

They hold when $|u|, |v|, |u+v| < 1$. The Euler constant cancels completely.

Theorem 5.2 (Finite triangular reductions). *For $n, m \geq 0$, $F_{n,m}^-(a)$ is a finite linear combination of 1 and $\gamma_j(a)$, $0 \leq j \leq n+m+1$. Its leading term is*

$$\frac{n+m+2}{(n+1)(m+1)} \gamma_{n+m+1}(a). \quad (5.5)$$

Similarly, $F_{n,m}^+(a)$ is a finite linear combination of 1, $\gamma_j(a)$, and $\gamma_j(b)$, with leading terms

$$\frac{\gamma_{n+m+1}(a)}{n+1} + \frac{\gamma_{n+m+1}(b)}{m+1}. \quad (5.6)$$

All coefficients belong to $\mathbb{Q}[\zeta(2), \dots, \zeta(n+m+2)]$. The terms with index $n+m$ are absent in both orientations. The formulas are constructive through (4.4), (5.2), (5.3), and (5.4).

Proof. The target coefficient has total degree $n+m+2$. Only finitely many terms of each displayed convergent germ can contribute. The highest Stieltjes term in B_- comes from $-(-1)^{n+m+1} \gamma_{n+m+1}(a)(u+v)^{n+m+2}/(n+m+1)!$. Multiply its coefficient by $(-1)^{n+m} n! m!$ to obtain (5.5). In (5.2), the factors v and u give the two binomial coefficients leading to (5.6). There are no degree-one terms in E, T, U , which proves the missing-index assertion. Equations (5.3)–(5.4) prove the coefficient ring statement without any numerical relation search. $\square$

5.1 Low-order identities

Here and below $\zeta_r = \zeta(r)$; in the following formulas all unmarked γ_j on the right are evaluated at a :

$$F_{0,0}^- = 2\gamma_1 + \zeta_2, \quad (5.7)$$

$$F_{1,0}^- = \frac{3}{2}\gamma_2 - \zeta_2\gamma_0 - \zeta_3, \quad (5.8)$$

$$F_{1,1}^- = \gamma_3 - 2\zeta_2\gamma_1 + 2\zeta_3\gamma_0 + \frac{1}{4}\zeta_4, \quad (5.9)$$

$$F_{2,0}^- = \frac{4}{3}\gamma_3 - 2\zeta_2\gamma_1 + 2\zeta_3\gamma_0 + 2\zeta_4. \quad (5.10)$$

In (5.9) we used $\zeta_2^2 = \frac{5}{2}\zeta_4$. The reflected identities are symmetric in n, m . For the other orientation,

$$F_{0,0}^+(a) = \gamma_1(a) + \gamma_1(b) - 2\zeta_2, \quad (5.11)$$

$$F_{1,0}^+(a) = \frac{1}{2}\gamma_2(a) + \gamma_2(b) + \zeta_2(\gamma_0(a) + \gamma_0(b)) - \zeta_3, \quad (5.12)$$

$$\begin{aligned} F_{1,1}^+(a) &= \frac{\gamma_3(a) + \gamma_3(b)}{2} + 2\zeta_2(\gamma_1(a) + \gamma_1(b)) \\ &\quad + \zeta_3(\gamma_0(a) + \gamma_0(b)) - \frac{7}{2}\zeta_4. \end{aligned} \quad (5.13)$$

One has $F_{n,m}^+(a) = F_{m,n}^+(b)$, rather than symmetry at a fixed a .

Corollary 5.3 (One digamma factor, every Stieltjes index). *For $n \geq 0$,*

$$\begin{aligned} F_{n,0}^-(a) &= \frac{n+2}{n+1}\gamma_{n+1}(a) + \sum_{j=1}^n (-1)^j \frac{n!}{(n-j)!} \zeta(j+1)\gamma_{n-j}(a) \\ &\quad + (-1)^n n! \zeta(n+2). \end{aligned} \quad (5.14)$$

Proof. Differentiate (5.3) in v at zero to get $E(u, v) = 1 - v \sum_{j \geq 1} \zeta(j+1)u^j + O(v^2)$. Insert this in (4.4), extract $[u^{n+1}v]$, and apply Theorem 4.2. This also independently verifies the entire first column of the coefficient table. $\square$

5.2 A polylogarithm order-derivative identity

Put $A = \gamma + \log(2\pi)$ and $z = e^{2\pi ia}$. Directly differentiating (4.5) at $u = v = 0$ yields

$$2 \operatorname{Re} \partial_s^2 \operatorname{Li}_s(z) \Big|_{s=0} - 4A \operatorname{Re} \partial_s \operatorname{Li}_s(z) \Big|_{s=0} = \gamma_1(a) + \gamma_1(b) + A^2 - \frac{\pi^2}{12}. \quad (5.15)$$

Indeed $\operatorname{Li}_0(z) + \operatorname{Li}_0(z^{-1}) = -1$, and the phase derivatives contribute $-\pi^2/4$ to $F_{0,0}^+$. Equation (5.11) then gives (5.15). This identity is also a useful independent normalization check on the two-variable germ.

For example, writing $\gamma_1 = \gamma_1(1)$, one obtains

$$F_{0,0}^+(\tfrac{1}{2}) = 2\gamma_1 - 4\gamma \log 2 - 2\log^2 2 - \frac{\pi^2}{3}, \quad (5.16)$$

$$F_{0,0}^+(\tfrac{1}{3}) = 2\gamma_1 - 3\gamma \log 3 - \frac{3}{2}\log^2 3 - \frac{\pi^2}{3}, \quad (5.17)$$

$$F_{0,0}^+(\tfrac{1}{4}) = 2\gamma_1 - 6\gamma \log 2 - 7\log^2 2 - \frac{\pi^2}{3}. \quad (5.18)$$

They follow either from Hurwitz distribution at the indicated moduli or from the root-of-unity polylogarithms in (4.5). No arithmetic independence claim is involved.

6 The periodic distribution algebra

The preceding integrals deliberately exclude the coincident shift. To describe convolution on the whole circle, including that point, one must retain distributions supported at zero. Let δ_0 be the periodic Dirac distribution, let 1 denote the constant distribution, and put

$$I = \delta_0 - 1. \quad (6.1)$$

On the space of distributions with mean zero, I is the convolution identity. It is not the identity on all periodic distributions.

For a smooth periodic test function ϕ , define

$$\langle Q_n, \phi \rangle = \int_0^1 \frac{\log^n x}{x} (\phi(x) - \phi(0)) dx + \int_0^1 \gamma_n(1+x) \phi(x) dx. \quad (6.2)$$

Thus Q_n is the scale-one finite part of γ_n . Its mean is zero: the first integral vanishes for $\phi = 1$, and $\int_1^2 \zeta(1+u, x) dx = 1/u$ implies $\int_1^2 \gamma_n(x) dx = 0$.

Let $Z(u)$ be the meromorphic distribution obtained by continuing $\zeta(1+u, x)$ from $\operatorname{Re} u < 0$. The singular monomial x^{-1-u} has residue $-\delta_0$, while the regular shifted Hurwitz term has residue 1. Consequently

$$Z(u) = -\frac{I}{u} + \mathcal{G}(u), \quad \mathcal{G}(u) = \sum_{n \geq 0} \frac{(-1)^n}{n!} Q_n u^n. \quad (6.3)$$

The series is holomorphic near zero in the distribution sense. A useful distinction is that the pointwise holomorphic function $H(u, x)$ in (2.2), when continued as a distribution, still has residue $-\delta_0$. Pointwise pole subtraction and spatial continuation do not remove the same pole.

Theorem 6.1 (A polynomial convolution algebra). *Let Y_j be the complete exponential Bell polynomials, defined by*

$$\exp \left(\sum_{j \geq 1} x_j \frac{u^j}{j!} \right) = \sum_{j \geq 0} Y_j(x_1, \dots, x_j) \frac{u^j}{j!}.$$

Interpret multiplication as periodic convolution and the scalar unit as I . Then

$$Q_n = \frac{(-1)^{n+1}}{n+1} Y_{n+1}(-Q_0, 1!\zeta(2)I, 2!\zeta(3)I, \dots, n!\zeta(n+1)I). \quad (6.4)$$

For $n = 0$ the list consists only of $-Q_0$. In particular,

$$Q_0 * Q_0 = 2Q_1 - \zeta(2)I, \quad (6.5)$$

$$Q_0 * Q_0 * Q_0 = 3Q_2 - 3\zeta(2)Q_0 + 2\zeta(3)I. \quad (6.6)$$

The algebra $\operatorname{span}_{\mathbb{C}}\{I, Q_0, Q_1, \dots\}$ is isomorphic to $\mathbb{C}[X]$ under $X \mapsto Q_0$.

Proof. Define P_u by its Fourier coefficients

$$\hat{P}_u(k) = \begin{cases} (2\pi i k)^u, & k \neq 0, \\ 0, & k = 0. \end{cases}$$

On compact parameter sets these coefficients have at most polynomial growth in k . Pairing with the rapidly decreasing coefficients of a smooth test function therefore proves that P_u is an entire distribution family. Fourier multiplication gives

$$P_0 = I, \quad P_u * P_v = P_{u+v}, \quad Z(u) = -\frac{\Gamma(1-u)}{u} P_u.$$

The last equality follows first from Lemma 2.1, then by meromorphic continuation. Comparing the constant term at $u = 0$ gives

$$\widehat{Q_0}(k) = -\gamma - \log(2\pi i k), \quad k \neq 0. \quad (6.7)$$

The classical expansion $\log \Gamma(1 - u) = \gamma u + \sum_{j \geq 2} \zeta(j) u^j / j$ now yields the distribution identity

$$u\mathcal{G}(u) = \mathbf{l} - \exp_* \left(-uQ_0 + \sum_{j \geq 2} \frac{\zeta(j)}{j} u^j \mathbf{l} \right), \quad |u| < 1. \quad (6.8)$$

Here the convolution exponential is justified by the Fourier family already constructed; no convergence assertion about arbitrary distribution exponentials is needed. Extracting coefficients proves (6.4).

The leading term of Q_n is $Q_0^{*(n+1)} / (n+1)$, so the indicated span is precisely the algebra generated by Q_0 and $\mathbf{l}$. If a polynomial p satisfies $p(Q_0) = 0$, its nonzero Fourier coefficients give $p(-\gamma - \log(2\pi i k)) = 0$ for every positive integer k . These are infinitely many distinct arguments, so p is the zero polynomial. This proves injectivity and the claimed isomorphism. $\square$

This functional algebraic independence does not imply rational or algebraic independence of any numerical special values. The proof uses the distinct Fourier frequencies, not unproved arithmetic assertions.

Proposition 6.2 (The full binary generating identity). *With $w = u + v$ and E as in (4.3),*

$$\mathcal{G}(u) * \mathcal{G}(v) = \frac{u\mathcal{G}(u) + v\mathcal{G}(v) - wE(u, v)\mathcal{G}(w) + (E(u, v) - 1)\mathbf{l}}{uv}. \quad (6.9)$$

The quotient is removable on both axes. Its restriction to $0 < a < 1$ is the finite-part germ $\mathcal{R}_-(u, v; a)$.

Proof. Multiply $u\mathcal{G}(u) = \mathbf{l} - \Gamma(1 - u)P_u$ and $v\mathcal{G}(v) = \mathbf{l} - \Gamma(1 - v)P_v$ using the semigroup law for P . Since $\Gamma(1 - u)\Gamma(1 - v) = E(u, v)\Gamma(1 - w)$, expansion gives (6.9). Each Q_n is smooth away from zero. At a nonzero shift the singular supports in the integral defining convolution are disjoint. A partition of unity around the two singularities therefore identifies its value with the separate cutoff finite parts in Definition 4.1. This proves the restriction assertion. $\square$

For clarity, the complete first reflected identity is

$$Q_0 * Q_0 = 2Q_1 + \zeta(2) \mathbf{l} - \zeta(2)\delta_0. \quad (6.10)$$

Let $\check{Q}(a) = Q(-a)$ denote reflection of a periodic distribution. The corresponding complete circular identity is

$$\check{Q}_0 * Q_0 = Q_1 + \check{Q}_1 - 2\zeta(2) \mathbf{l} + 2\zeta(2)\delta_0. \quad (6.11)$$

For $k \neq 0$, put $c_k = \gamma + \log(2\pi|k|)$. Its two sides have Fourier coefficient $c_k^2 + \pi^2/4$, as follows from (6.7) and $\widehat{Q_1}(k) = ((\gamma + \log(2\pi i k))^2 + \zeta(2))/2$. Their zero coefficients also agree. Omitting the Dirac terms in these global identities would contradict the zero mean; omitting them in the interior formulas is legitimate because $a \neq 0$.

Proposition 6.3 (The joint algebra of both orientations). *Let*

$$X = \frac{Q_0 + \check{Q}_0}{2}, \quad J = \frac{\check{Q}_0 - Q_0}{\pi i}.$$

The finite convolution algebra generated by all Q_n and their reflections is

$$\mathbb{C}[X, J] / (J^2 - 1) \cong \mathbb{C}[X] \times \mathbb{C}[X], \quad (6.12)$$

under $X \mapsto X$, $J \mapsto J$, and $\mathbf{l} \mapsto \mathbf{l}$.

Proof. At a nonzero frequency k , the Fourier coefficients are $\widehat{X}(k) = -\gamma - \log(2\pi|k|)$ and $\widehat{J}(k) = \operatorname{sgn} k$. Both means are zero; hence $J * J = I$. Conversely $Q_0 = X - (\pi i/2)J$ and $Q_0 = X + (\pi i/2)J$, and all other generators are Bell polynomials in these two. If $p(X) + J * q(X) = 0$, its positive and negative Fourier coefficients imply that both $p + q$ and $p - q$ vanish at every $-\gamma - \log(2\pi n)$, $n \geq 1$. Thus $p = q = 0$, proving that $J^2 - 1$ is the full ideal of relations. Evaluation at $J = 1$ and $J = -1$ gives the last isomorphism. This assertion concerns finite polynomials, with no topological closure implicit. $\square$

7 Differentiation of finite parts and polygamma products

Differentiating a finite-part distribution is not always the same operation as taking the raw finite part of the pointwise derivative. We specify the latter without an implicit choice of additive constants. For integers $m, n \geq 0$, let $T_{m,n} = \operatorname{FP}[x^{-m-1} \log^n x]$ mean

$$\begin{aligned} \langle T_{m,n}, \phi \rangle &= \int_0^1 x^{-m-1} \log^n x \left(\phi(x) - \sum_{j=0}^m \frac{\phi^{(j)}(0)}{j!} x^j \right) dx \\ &\quad + \sum_{j=0}^{m-1} \frac{\phi^{(j)}(0)}{j!} \frac{(-1)^n n!}{(j-m)^{n+1}}. \end{aligned} \quad (7.1)$$

The second sum is empty when $m = 0$. This is exactly the constant term of the cutoff integral at scale one after all divergent powers and logarithms have been removed. Apply (7.1) to the singular polynomial in the derivative of (2.3), and integrate its smooth remainder ordinarily. Denote the resulting distribution by $P_{n,k} = \operatorname{FP} \gamma_n^{(k)}$; set $P_{n,0} = Q_n$.

Let e_j denote an elementary symmetric polynomial and $H_k^{(j)} = \sum_{r=1}^k r^{-j}$, with $H_k = H_k^{(1)}$.

Theorem 7.1 (The derivative correction). *For every $n \geq 0$ and $k \geq 1$,*

$$P_{n,k} = D^k Q_n + c_{n,k} \delta_0^{(k)}, \quad c_{n,k} = (-1)^n n! e_{n+1} \left(1, \frac{1}{2}, \dots, \frac{1}{k} \right). \quad (7.2)$$

In particular, $c_{0,k} = H_k$, and $c_{n,k} = 0$ when $n \geq k$.

Proof. The meromorphic continuation $X_m(u)$ of x^{-m-1-u} is obtained by subtracting the test-function Taylor polynomial in (7.1). Integration of its monomials gives

$$\langle X_m(u), \phi \rangle = \int_0^1 x^{-m-1-u} (\phi - \operatorname{Taylor}_m \phi) dx + \sum_{j=0}^m \frac{\phi^{(j)}(0)}{j!(j-m-u)}.$$

Using $\langle \delta_0^{(m)}, \phi \rangle = (-1)^m \phi^{(m)}(0)$, its Laurent expansion is

$$X_m(u) = -\frac{(-1)^m}{m!u} \delta_0^{(m)} + \sum_{n \geq 0} \frac{(-1)^n}{n!} T_{m,n} u^n. \quad (7.3)$$

On the half-plane $\operatorname{Re} u < -k$, the derivatives through order $k-1$ of $\zeta(1+u, x)$ match at the periodic endpoints and its k th derivative is integrable. Thus ordinary integration by parts and meromorphic continuation give

$$D^k Z(u) = (-1)^k (1+u)_k [X_k(u) + \zeta(k+1+u, 1+x)].$$

Here $(1+u)_k = (1+u) \cdots (k+u)$ is the rising factorial. Put $E_k(u) = (1+u)_k/k! = \prod_{j=1}^k (1+u/j)$. Equation (7.3) shows that the last display equals

$$-\frac{E_k(u)}{u} \delta_0^{(k)} + \sum_{n \geq 0} \frac{(-1)^n}{n!} P_{n,k} u^n.$$

On the other hand, differentiating (6.3) gives $-\delta_0^{(k)}/u + \sum (-1)^n D^k Q_n u^n/n!$. Their difference is

$$\sum_{n \geq 0} \frac{(-1)^n}{n!} (P_{n,k} - D^k Q_n) u^n = \frac{E_k(u) - 1}{u} \delta_0^{(k)}.$$

Coefficient comparison proves (7.2), including the vanishing assertion. $\square$

For example,

$$P_{0,1} = DQ_0 + \delta'_0, \quad P_{0,2} = D^2Q_0 + \tfrac{3}{2}\delta''_0, \quad P_{1,2} = D^2Q_1 - \tfrac{1}{2}\delta''_0.$$

These examples illustrate that the correction depends on both indices.

Corollary 7.2 (Every-order digamma–polygamma integral). *For $k \geq 1$ and $0 < a < 1$, with the raw finite part just defined,*

$$\begin{aligned} \text{FP} \int_0^1 \psi(x) \psi^{(k)}(\{a-x\}) dx &= 2\gamma_1^{(k)}(a) - H_k \psi^{(k)}(a) \\ &= (-1)^{k+1} k! [2\zeta'(k+1, a) + H_k \zeta(k+1, a)]. \end{aligned} \quad (7.4)$$

Proof. Convolve $P_{0,k} = D^k Q_0 + H_k \delta_0^{(k)}$ with Q_0 and use (6.10). Away from zero the result is $2\gamma_1^{(k)} + H_k \gamma_0^{(k)}$. The two minus signs from $\gamma_0 = -\psi$ cancel in the product. For the second equality, differentiate the Laurent series and use

$$\gamma_1^{(k)}(a) = (-1)^{k+1} k! [\zeta'(k+1, a) + H_k \zeta(k+1, a)], \quad \psi^{(k)}(a) = (-1)^{k+1} k! \zeta(k+1, a). \quad (7.5)$$

The first relation follows from the coefficient of u in $\partial_a^k \zeta(1+u, a) = (-1)^k (1+u)_k \zeta(k+1+u, a)$. $\square$

The full distribution identity behind (7.4) is

$$Q_0 * P_{0,k} = 2P_{1,k} + H_k P_{0,k} - (H_k^{(2)} + \zeta(2)) \delta_0^{(k)}. \quad (7.6)$$

It follows by substituting $c_{1,k} = -(H_k^2 - H_k^{(2)})/2$ into Theorem 7.1.

For $k = 1$, the same finite part can be written particularly concretely:

$$\begin{aligned} \lim_{\epsilon \downarrow 0} \left[\int_{\epsilon}^{a-\epsilon} \psi(x) \psi'(a-x) dx + \int_a^1 \psi(x) \psi'(1+a-x) dx \right. \\ \left. - \frac{\psi(a)}{\epsilon} - 2\psi'(a) \log \epsilon \right] = 2\zeta'(2, a) + \zeta(2, a). \end{aligned} \quad (7.7)$$

At $a = 1/2$ this becomes

$$6\zeta'(2) + \frac{4\pi^2}{3} \log 2 + \frac{\pi^2}{2} = 8.43096395979 \dots \quad (7.8)$$

The power counterterm in (7.7) is indispensable; the logarithmic counterterms of Definition 4.1 alone do not apply to a differentiated factor.

More generally, put $c_{n,0} = 0$. Every pair of raw derivatives obeys

$$P_{n,k} * P_{m,\ell} = D^{k+\ell}(\mathbf{Q}_n * \mathbf{Q}_m) + c_{m,\ell} D^{k+\ell} \mathbf{Q}_n + c_{n,k} D^{k+\ell} \mathbf{Q}_m + c_{n,k} c_{m,\ell} \delta_0^{(k+\ell)}. \quad (7.9)$$

This identity, together with (6.9), supplies a finite algorithm for all the corresponding separated-singularity integrals.

8 Ordinary log-Gamma correlations and antiderivatives

Let

$$L = \log(2\pi), \quad A = \gamma + L, \quad g(x) = \log \Gamma(x) - \frac{L}{2} = \zeta'(0, x). \quad (8.1)$$

Lerch's identity gives the last equality. The function g has mean zero and belongs to $L^2(0, 1)$. Define the ordinary, absolutely convergent integrals

$$R(a) = \int_0^1 \log \Gamma(x) \log \Gamma(\{x + a\}) dx, \quad (8.2)$$

$$S(a) = \int_0^1 \log \Gamma(x) \log \Gamma(\{a - x\}) dx. \quad (8.3)$$

At the coincident shifts the product grows only as $\log^2 x$, so the definitions extend continuously to the whole circle.

Proposition 8.1 (Two ordinary correlation evaluations). *For $0 < a < 1$, put $\mathcal{L}_w(a) = (\text{Li}_w(e^{2\pi i a}) + \text{Li}_w(e^{-2\pi i a}))/2$. Then*

$$R(a) = \frac{L^2}{4} + \frac{1}{2\pi^2} \left[(A - \partial_w)^2 + \frac{\pi^2}{4} \right] \mathcal{L}_w(a) \Big|_{w=2}, \quad (8.4)$$

and

$$S(a) = \frac{L^2}{4} + \zeta''(-1, a) + 2\zeta'(-1, a) + (2 - \zeta(2))\zeta(-1, a). \quad (8.5)$$

Their interior second derivatives satisfy

$$R''(a) = \frac{\pi^2}{3} - \gamma_1(a) - \gamma_1(1 - a), \quad S''(a) = 2\gamma_1(a) + \zeta(2). \quad (8.6)$$

Proof. Differentiate the Fourier coefficient in Lemma 2.1 at $s = 0$. For $n \geq 1$,

$$\widehat{g}(n) = \frac{A + \log n + \pi i/2}{2\pi i n}.$$

The coefficients of its circular autocorrelation are their squared absolute values. Hence the absolutely and uniformly convergent series is

$$R(a) = \frac{L^2}{4} + \sum_{n \geq 1} \frac{(A + \log n)^2 + \pi^2/4}{2\pi^2 n^2} \cos(2\pi n a). \quad (8.7)$$

This is (8.4); order differentiation is legitimate because $\sum n^{-2} \log^2 n < \infty$. The symmetric polylogarithm average, rather than taking a real part at complex w , preserves holomorphy during differentiation.

For the reflected formula, apply $\partial_s \partial_t$ at $s = t = 0$ to (3.1). The coefficient of st in $\Gamma(1-s)\Gamma(1-t)/\Gamma(2-s-t)$ is $2 - \zeta(2)$, and its two first coefficients equal 1. This gives (8.5).

As a periodic distribution $Dg = -Q_0$. This follows either from the Fourier coefficients or by integrating the local behavior $g(x) = -\log x - L/2 + O(x)$ by parts with the scale-one cutoff. Thus $D^2(g * g) = Q_0 * Q_0$, while $D^2(\check{g} * g) = -Q_0 * Q_0$. Restrict (6.10)–(6.11) to the interior to obtain (8.6). These restrictions are smooth there, which also justifies interpreting the derivatives classically. $\square$

In particular,

$$R(0) = \frac{L^2}{4} + \frac{(A^2 + \pi^2/4)\zeta(2) - 2A\zeta'(2) + \zeta''(2)}{2\pi^2}. \quad (8.8)$$

This is an ordinary second moment. It should not be confused with the finite parts at order one.

8.1 A critical-order integral

The same master formula gives an unexpectedly short evaluation at a different pair of orders:

$$\int_0^1 \zeta(s, x) \zeta(1-s, \{x+a\}) dx = -\log(2 \sin \pi a) + \pi(a - \tfrac{1}{2}) \cot \pi s, \quad 0 < \operatorname{Re} s < 1. \quad (8.9)$$

Indeed substitute $t = 1 - s$ in (3.2), use $\operatorname{Li}_1(z) = -\log(1-z)$ and $\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$, and retain the arguments of $1 - e^{\pm 2\pi i a}$ continuously for $0 < a < 1$. In particular,

$$\int_0^1 \zeta(s, x) \zeta(1-s, \{x + \tfrac{1}{2}\}) dx = -\log 2. \quad (8.10)$$

It is independent of s throughout the strip. This is an explicit consequence of the classical convolution formula, not a separate novelty claim.

9 An exact cusp expansion and strict convexity

In this section γ_1 without an argument means $\gamma_1(1)$. For $m \geq 1$ set

$$B_m = H_{2m} \zeta(2m+1) + \zeta'(2m+1). \quad (9.1)$$

These constants also occur in the even parameter derivatives of γ_1 at one, by (7.5).

Theorem 9.1 (Convergent cusp expansion). *For every $0 < a < 1$,*

$$\begin{aligned} R(0) - R(a) &= \frac{a}{2} \left(\log^2 a - 2 \log a + 2 + \frac{\pi^2}{3} \right) + (\gamma_1 - \zeta(2))a^2 \\ &\quad - \sum_{m \geq 1} \frac{B_m a^{2m+2}}{(m+1)(2m+1)}. \end{aligned} \quad (9.2)$$

The final series converges locally uniformly, with all its derivatives, for $|a| < 1$. Every B_m is strictly positive.

Proof. Write $\theta = 2\pi a$. The symmetric form of the local polylogarithm expansion [3, 25.12.12] is

$$\frac{\text{Li}_w(e^{i\theta}) + \text{Li}_w(e^{-i\theta})}{2} = J(w)\theta^{w-1} + \sum_{j \geq 0} \frac{(-1)^j \zeta(w-2j)}{(2j)!} \theta^{2j}, \quad J(w) = \frac{\pi}{2\Gamma(w) \cos(\pi w/2)}. \quad (9.3)$$

Near $w = 2$ every term displayed is holomorphic, and the series converges normally for $|\theta| < 2\pi$. Thus the differential operator in (8.4) may be applied termwise. Direct logarithmic differentiation gives

$$J(2) = -\frac{\pi}{2}, \quad \frac{J'(2)}{J(2)} = \gamma - 1, \quad \left(\frac{J'}{J}\right)'(2) = 1 + \frac{\pi^2}{12}.$$

The contribution of $J(w)\theta^{w-1}$ to $R(a)$ is therefore $-a(\log^2 a - 2\log a + 2 + \pi^2/3)/2$. The $j = 0$ term is $R(0) - L^2/4$.

For $j = 1$, the functional equation of the Riemann zeta function [4, 25.4.2] gives

$$\zeta''(0) = \gamma_1 + \frac{\gamma^2}{2} - \frac{\pi^2}{24} - \frac{L^2}{2}. \quad (9.4)$$

For completeness, expand $\zeta(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) \zeta(1-s)$ through degree two at zero; the Laurent part is $\zeta(1-s) = -1/s + \gamma + \gamma_1 s + O(s^2)$, which gives (9.4). Together with $\zeta(0) = -1/2$ and $\zeta'(0) = -L/2$, this evaluates the quadratic coefficient of $R(a)$ as $\zeta(2) - \gamma_1$.

For $j = m+1 \geq 2$, the trivial zero at $-2m$ removes the undifferentiated zeta value. Expanding the same functional equation there gives

$$\zeta'(-2m) = \frac{(-1)^m (2m)!}{2(2\pi)^{2m}} \zeta(2m+1), \quad (9.5)$$

$$\zeta''(-2m) - 2A\zeta'(-2m) = \frac{(-1)^{m+1} (2m)!}{(2\pi)^{2m}} B_m. \quad (9.6)$$

One can see the second equality directly by writing $\zeta(s) = \sin(\pi s/2) F(s)$ near $s = -2m$. Then $\zeta''(-2m) = 2\zeta'(-2m)F'(-2m)/F(-2m)$, and

$$\frac{F'(-2m)}{F(-2m)} = A - H_{2m} - \frac{\zeta'(2m+1)}{\zeta(2m+1)}.$$

Substitution into (9.3) gives exactly the series coefficient in (9.2).

Finally $H_{2m}\zeta(2m+1) > 1$, whereas

$$0 < -\zeta'(2m+1) \leq -\zeta'(3) = \sum_{n \geq 2} \frac{\log n}{n^3} < \sum_{n \geq 2} \frac{n-1}{n^3} = \zeta(2) - \zeta(3) < 1.$$

The final inequality follows, for example, from $\zeta(2) < 2$ and $\zeta(3) > 1$. Hence $B_m > 0$. Also $B_m < H_{2m}\zeta(3) = O(1 + \log m)$, which independently proves the normal convergence and all local derivative convergence asserted. $\square$

For a useful explicit error bound, retain the terms $1 \leq m \leq M$, with $M \geq 0$. The magnitude of the omitted series in (9.2) is at most

$$\frac{\zeta(3)a^{2M+4}}{(M+2)(1-a^2)}. \quad (9.7)$$

Indeed $B_m < 2m\zeta(3)$ and $2m/((m+1)(2m+1)) < 1/(m+1)$. Since $R(a) = R(1-a)$, numerical evaluation may always use $0 < a \leq 1/2$, where this geometric bound is effective.

Corollary 9.2 (Positive even derivatives and the unique minimum). *For every integer $k \geq 1$ and $0 < a < 1$,*

$$R^{(2k)}(a) > 0. \quad (9.8)$$

Every odd derivative is strictly increasing and vanishes at $a = 1/2$. Every even derivative has its unique minimum there. In particular R is strictly convex, strictly decreasing on $(0, 1/2)$, and strictly increasing on $(1/2, 1)$.

Proof. Differentiating (9.2) twice gives

$$R''(a) = -\frac{\log a}{a} + 2(\zeta(2) - \gamma_1) + 2 \sum_{m \geq 1} B_m a^{2m}. \quad (9.9)$$

We record a simple rigorous bound on the only constant whose sign is needed. By the defining Stieltjes limit,

$$\gamma_1 = \lim_{N \rightarrow \infty} \left(\sum_{n=1}^N \frac{\log n}{n} - \frac{1}{2} \log^2 N \right).$$

The function $\log x/x$ decreases for $x \geq 3$. Comparing the terms $n \geq 4$ with its integral from 3 gives

$$\gamma_1 \leq \frac{\log 2}{2} + \frac{\log 3}{3} - \frac{1}{2} \log^2 3 < 1 < \zeta(2).$$

All terms in (9.9) are therefore positive. For $k \geq 2$, put $j = 2k - 2$. The logarithmic term obeys

$$\frac{d^j}{da^j} \left(-\frac{\log a}{a} \right) = j! a^{-j-1} (H_j - \log a) > 0.$$

Every surviving derivative of the even-power series is nonnegative, proving (9.8). The symmetry $R(a) = R(1-a)$ makes each odd derivative vanish at the midpoint. Its next derivative is positive, so it is strictly increasing. Applying this to the derivatives of each even derivative proves their unique midpoint minima. $\square$

Corollary 9.3 (Sharp quadratic lower bound and reflection inequality). *Put*

$$M = R''(\tfrac{1}{2}) = \frac{\pi^2}{3} - 2\gamma_1(\tfrac{1}{2}) = \frac{\pi^2}{3} - 2\gamma_1 + 4\gamma \log 2 + 2 \log^2 2 = 5.99678749530633590836 \dots \quad (9.10)$$

For $0 \leq a \leq 1$,

$$R(a) \geq R(\tfrac{1}{2}) + \frac{M}{2} (a - \tfrac{1}{2})^2. \quad (9.11)$$

Equality occurs only at $a = 1/2$, and the coefficient $M/2$ is optimal. For $0 < a < 1$ one also has

$$\gamma_1(a) + \gamma_1(1-a) \leq 2\gamma_1(\tfrac{1}{2}), \quad (9.12)$$

with equality only at the midpoint.

Proof. The preceding corollary gives $R''(a) \geq M$ with strict inequality off the midpoint. Integrate twice from $1/2$, where $R' = 0$, and extend to the endpoints by continuity. Taylor expansion at the midpoint proves optimality of the coefficient. Equation (8.6) turns the same curvature bound into (9.12). $\square$

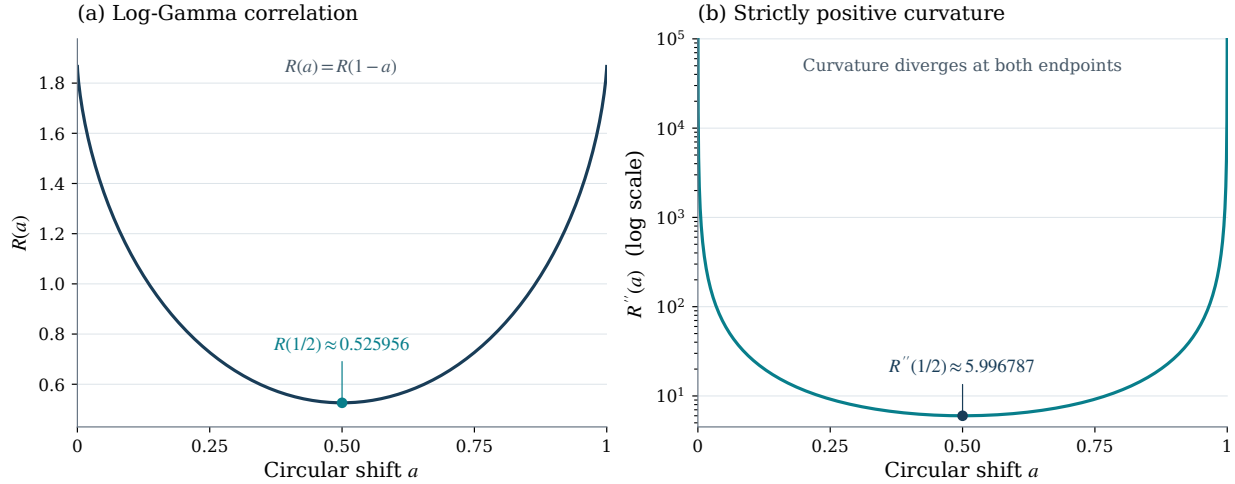


Figure 1: The ordinary circular log-Gamma correlation and its curvature. The plot uses the convergent expansion (9.2) and the series (9.9); the analytic statements and strict inequalities follow from the proofs, not from the plot. Both minima occur at $a = 1/2$.

The leading cusp has the particularly simple form

$$R(0) - R(a) \sim \frac{a}{2} \log^2(1/a) \quad (a \downarrow 0). \quad (9.13)$$

Consequently the periodic function R is Hölder continuous of every exponent $\alpha < 1$ but is not Lipschitz continuous. Translation in $L^2(\mathbb{T})$ gives the equivalent sharp norm law

$$\|\log \Gamma(\{\cdot + a\}) - \log \Gamma(\cdot)\|_2^2 = 2(R(0) - R(a)) \sim a \log^2(1/a). \quad (9.14)$$

Thus the cusp measures the precise loss of smoothness under small translations of log-Gamma.

9.1 Recovery of a known Dilcher series

Setting $R'(1/2) = 0$ in the differentiated expansion gives

$$\gamma_1 + \frac{1}{2} \log^2 2 = \sum_{m \geq 1} \frac{H_{2m} \zeta(2m+1) + \zeta'(2m+1)}{(2m+1)4^m}. \quad (9.15)$$

Every summand is positive. The tail after $m = M \geq 0$ is less than $\zeta(3)/(3 \cdot 4^M)$, by $H_{2m} < 2m+1$. In particular $\gamma_1 > -(\log 2)^2/2$ follows without a decimal evaluation.

This identity is not claimed as new. Brede [10, printed p. 16] attributes the equivalent formula

$$\gamma_1 = -\frac{1}{2} \log^2(3/2) + \sum_{m \geq 1} \frac{H_{2m}(\zeta(2m+1) - 1) + \zeta'(2m+1)}{(2m+1)4^m} \quad (9.16)$$

to Dilcher [9]. Equivalence follows from

$$\sum_{m \geq 1} \frac{H_{2m}}{(2m+1)4^m} = \frac{1}{2} (\log^2 2 - \log^2(3/2)).$$

For a direct verification, take the even part of $\sum_{n \geq 0} H_n x^n = -\log(1-x)/(1-x)$, integrate from 0 to $1/2$, and divide by $1/2$. The source chain is explicit because the present audit verified the displayed identity in Brede's accessible dissertation and the primary Dilcher article's bibliographic record, without access to the full Dilcher text.

10 All negative-integer jets and Stieltjes antiderivatives

The log-Gamma case is the first member of a larger exact regularity theorem. For integers $p, r \geq 0$ set

$$Z_{p,r}(x) = \zeta^{(r)}(-p, x), \quad R_{p,r}(a) = \int_0^1 Z_{p,r}(x) Z_{p,r}(\{x+a\}) dx. \quad (10.1)$$

All these functions are in $L^2(0, 1)$ and have mean zero. Their correlations are real and even. Write $H_0^{(j)} = 0$ and define the monic polynomial

$$P_{p,r}(z) = r![u^r] \exp \left(zu + \sum_{j \geq 2} \frac{\zeta(j) - H_p^{(j)}}{j} u^j \right). \quad (10.2)$$

Only terms through degree r in this exponential contribute. Put

$$Q_{p,r}(y) = P_{p,r}(y + A - H_p + \pi i/2) P_{p,r}(y + A - H_p - \pi i/2). \quad (10.3)$$

Its coefficients are real and its degree is $2r$.

Theorem 10.1 (Complete collision polynomial for every jet). *Define*

$$E_{p,r,k} = \frac{2(p!)^2}{(2\pi)^{2p+2}} \frac{(-1)^k (2\pi)^{2k}}{(2k)!} Q_{p,r}(-\partial_w) \zeta(w - 2k)|_{w=2p+2}, \quad k \geq 0, \quad (10.4)$$

and the holomorphic beta germ

$$K_p(u, v) = \frac{(-1)^{p+1}}{2} B(p+1-u, p+1-v) \frac{\cos(\pi(u-v)/2)}{\cos(\pi(u+v)/2)}. \quad (10.5)$$

Here $B(x, y) = \Gamma(x)\Gamma(y)/\Gamma(x+y)$. Then for every $0 < a < 1$,

$$R_{p,r}(a) = \sum_{k \geq 0} E_{p,r,k} a^{2k} + a^{2p+1} (r!)^2 [u^r v^r] \left(K_p(u, v) e^{-(u+v) \log a} \right). \quad (10.6)$$

The series converges normally for $|a| < 1$. The second term is a^{2p+1} times a polynomial of degree $2r$ in $\log a$, all of whose coefficients are determined by the finite germ in (10.5).

Proof. Differentiate the Hurwitz Fourier coefficient at $s = -p$. Since

$$\log \frac{\Gamma(p+1-u)}{p!} = (\gamma - H_p)u + \sum_{j \geq 2} \frac{\zeta(j) - H_p^{(j)}}{j} u^j,$$

the positive Fourier coefficients are

$$\hat{Z}_{p,r}(n) = \frac{p! e^{-\pi i(p+1)/2}}{(2\pi n)^{p+1}} P_{p,r}(\log n + A - H_p + \pi i/2), \quad n \geq 1. \quad (10.7)$$

Their integrals and parameter derivatives are justified on a sufficiently small spectral disk by majorants $C[1 + x^{p-\epsilon}(1 + |\log x|^r)]$, with fixed $0 < \epsilon < 1/2$. These include the bounded shifted Hurwitz remainder and belong to L^2 ; the resulting Fourier families also converge in L^2 . Squaring their absolute values shows

$$R_{p,r}(a) = \frac{2(p!)^2}{(2\pi)^{2p+2}} Q_{p,r}(-\partial_w) \mathcal{L}_w(a) \Big|_{w=2p+2}. \quad (10.8)$$

The even analytic terms in (9.3) give precisely (10.4). None of their zeta arguments is 1 at this even evaluation order, so there is no unpaired pole.

To compute the nonanalytic term without expanding many derivatives, symmetrize (3.2) in s, t before taking equal-order jets. This gives

$$\frac{\mathcal{K}_+(s, t; a) + \mathcal{K}_+(t, s; a)}{2} = 2\Gamma(1-s)\Gamma(1-t)(2\pi)^{s+t-2} \cos \frac{\pi(s-t)}{2} \mathcal{L}_{2-s-t}(a).$$

Its fractional-power coefficient is

$$\frac{\Gamma(1-s)\Gamma(1-t)\Gamma(s+t-1)}{2\pi} (\sin \pi s + \sin \pi t).$$

Substitute $s = -p + u, t = -p + v$. Euler reflection converts this coefficient to $K_p(u, v)$; the zero in the sum of sines cancels the apparent gamma pole. The power is $a^{2p+1-u-v}$. Taking $\partial_u^r \partial_v^r$ proves the collision term in (10.6). Normal convergence of the local polylogarithm expansion on compact parameter disks justifies every differentiation and the stated convergence. $\square$

For the first jet $r = 1$, the whole collision polynomial simplifies to

$$\begin{aligned} R_{p,1}(a) &= \sum_{k \geq 0} E_{p,1,k} a^{2k} + \frac{(-1)^{p+1} (p!)^2 a^{2p+1}}{2(2p+1)!} \\ &\quad \times \left[(\log a - H_{2p+1} + H_p)^2 + H_{2p+1}^{(2)} + \frac{\pi^2}{3} \right]. \end{aligned} \quad (10.9)$$

To check the coefficient explicitly, the first logarithmic derivative of K_p at zero is $H_{2p+1} - H_p$, and its mixed logarithmic derivative is $H_{2p+1}^{(2)} + \pi^2/3$. With $p = 0$, formula (10.9) recovers the nonanalytic part of Theorem 9.1.

For a higher example, $p = 0, r = 2$, let $T = 1 - \log a$. The complete collision contribution is

$$-\frac{a}{2} \left[T^4 + (6 + 8\zeta(2))T^2 + 8(1 - \zeta(3))T + 9 + 8\zeta(2) + 14\zeta(4) \right]. \quad (10.10)$$

It follows by extracting $[u^2 v^2]$ in the holomorphic germ, so its constants are exact and do not rely on a numerical fit.

Corollary 10.2 (Sharp regularity). *On $\mathbb{T}$, for $p, r \geq 0$,*

$$R_{p,r} \in C^{2p}, \quad R_{p,r} \notin C^{2p+1}. \quad (10.11)$$

For $r \geq 1$ one has $R_{p,r} \in C^{2p,\alpha}$ for every $0 < \alpha < 1$, but $R_{p,r} \notin C^{2p,1}$. For $r = 0$ it belongs to $C^{2p,1}$ and its next derivative has a jump at the seam. The exact Sobolev thresholds are

$$Z_{p,r} \in H^q(\mathbb{T}) \iff q < p + \frac{1}{2}, \quad R_{p,r} \in H^q(\mathbb{T}) \iff q < 2p + \frac{3}{2}. \quad (10.12)$$

Proof. Since $P_{p,r}$ is monic, (10.7) has magnitude asymptotic to $p!n^{-p-1}(\log n)^r/(2\pi)^{p+1}$. The correlation coefficients are its squared magnitudes. The weighted square-sum definition of H^q proves (10.12), including divergence at both threshold exponents.

The leading coefficient of the logarithmic collision polynomial is

$$K_p(0, 0) = \frac{(-1)^{p+1} (p!)^2}{2(2p+1)!}. \quad (10.13)$$

Indeed the factor $(r!)^2[u^r v^r]e^{-(u+v)\log a}$ has leading term $\log^{2r} a$. The analytic even series and all its relevant derivatives converge through the seam, whereas the collision term has a continuous $2p$ th derivative and

$$R_{p,r}^{(2p)}(a) - R_{p,r}^{(2p)}(0) \sim \frac{(-1)^{p+1}(p!)^2}{2} |a| \log^{2r} |a|. \quad (10.14)$$

For $r = 0$ the logarithmic factor means 1. This proves the failure of the next derivative and, when $r \geq 1$, the failure of Lipschitz continuity of the $2p$ th derivative. The elementary bound $h(1 + |\log h|^{2r}) = O(h^\alpha)$ for $\alpha < 1$, applied to the local collision polynomial, proves the asserted Hölder regularity. Away from the seam everything is real analytic. $\square$

10.1 Application to the normalized Stieltjes primitive

Use the manuscript's normalization

$$\mathcal{A}_n(x) = \frac{(-1)^{n+1}}{n+1} (\zeta^{(n+1)}(0, x) - \zeta^{(n+1)}(0)). \quad (10.15)$$

Then $\mathcal{A}_n(1) = 0$ and $\mathcal{A}'_n(x) = \gamma_n(x)$. The latter follows by taking the coefficient of s^{n+1} in $\partial_x \zeta(s, x) = -s\zeta(s+1, x)$; it is one of the existing antiderivative identities, included here to fix the normalization. After subtracting its mean, $\mathcal{A}_n$ is $(-1)^{n+1} Z_{0,n+1}/(n+1)$. Its circular covariance V_n therefore has the exact representation $V_n = R_{0,n+1}/(n+1)^2$ and

$$V_n(0) - V_n(a) \sim \frac{a}{2(n+1)^2} \log^{2n+2}(1/a). \quad (10.16)$$

The full logarithmic polynomial is given by Theorem 10.1, not only its leading coefficient. In particular every one of these primitive covariances is Hölder of every exponent below one and fails to be Lipschitz. This extends the antiderivative formulas to sharp analytic information about their translated products.

The balanced negative-polygamma combination

$$\Psi_{p+1}(x) = \frac{\zeta'(-p, x) + H_p \zeta(-p, x)}{p!}$$

has the similarly explicit collision contribution

$$\frac{(-1)^{p+1} a^{2p+1}}{2(2p+1)!} \left[(\log a - H_{2p+1})^2 + H_{2p+1}^{(2)} + \frac{\pi^2}{3} \right]. \quad (10.17)$$

Its analytic coefficients follow from (10.4) by replacing the first-jet polynomial z with $z + H_p$ and dividing by $(p!)^2$.

11 Positive moments in the Stieltjes derivative tower

The derivative identities admit a complementary nonlinear consequence. It comes from a positive measure, rather than a numerical search for linear relations. Let $a > 0$, $k \geq 1$, and $p = k + 1$. Define

$$f_n = \frac{(-1)^k}{k!} \gamma_n^{(k)}(a), \quad n \geq 0. \quad (11.1)$$

The parameter-derivative identity yields

$$\sum_{n \geq 0} f_n \frac{z^n}{n!} = \prod_{j=1}^k \left(1 - \frac{z}{j}\right) \zeta(p - z, a). \quad (11.2)$$

The right side is entire in z : the polynomial cancels the sole Hurwitz pole, at $z = k$. In particular, these are actual globally convergent Taylor series, not merely formal expansions.

Let $h_r = h_r(1, 1/2, \dots, 1/k)$ be the complete homogeneous symmetric polynomial, with $h_0 = 1$, so that

$$\prod_{j=1}^k (1 - z/j)^{-1} = \sum_{r \geq 0} h_r z^r \quad \text{near } z = 0.$$

The compensated coefficients are

$$M_n = \sum_{r=0}^n \frac{n!}{(n-r)!} h_r f_{n-r} = (-1)^n \zeta^{(n)}(p, a) = \sum_{j \geq 0} \frac{\log^n(a+j)}{(a+j)^p}. \quad (11.3)$$

The compensated generating function $\zeta(p-z, a)$ does have a pole at $z = k$; it should be distinguished from the entire function in (11.2).

Theorem 11.1 (Strict Hankel positivity and an exact determinant). *For every $a > 0$, real $p > 1$, and integer $d \geq 1$, define $M_n = (-1)^n \zeta^{(n)}(p, a)$. Then*

$$D_d(p, a) := \det[M_{i+j}]_{0 \leq i, j < d} = \sum_{0 \leq j_1 < \dots < j_d} \prod_{v=1}^d (a + j_v)^{-p} \prod_{1 \leq u < v \leq d} \log^2 \left(\frac{a + j_v}{a + j_u} \right) > 0. \quad (11.4)$$

Thus every finite Hankel moment matrix is strictly positive definite. At integer $p = k + 1$, (11.3) expresses all these strict inequalities as polynomial inequalities in Stieltjes parameter derivatives.

Proof. The positive discrete measure

$$\mu_{p,a} = \sum_{j \geq 0} (a+j)^{-p} \delta_{\log(a+j)}$$

has finite moments of every order and exactly the moments M_n . For a nonzero polynomial q , $\int q(t)^2 d\mu_{p,a}(t) > 0$, because infinitely many distinct support points cannot all be roots of q . This proves strict positive definiteness. For the determinant identity, restrict the measure to $0 \leq j \leq N$. Its Gram matrix factors as the product of a monomial evaluation matrix, the diagonal weights, and the transpose evaluation matrix. The finite Cauchy–Binet identity gives the sum of the squared Vandermonde minors displayed in (11.4). Let $N \rightarrow \infty$. The moments, hence determinants, converge absolutely; the positive tuple sums increase to the same limit. This proves the identity and strictness without a cancellation estimate. $\square$

For $a \geq 1$, this is a Stieltjes moment measure, supported on $[0, \infty)$. When $0 < a < 1$, its first support point is negative, so the correct general description is a Hamburger moment measure; odd moments need not be positive. The Hankel conclusions hold in both cases.

Corollary 11.2 (A strict nonlinear Stieltjes inequality). *For every $a > 0$ and integer $k \geq 1$,*

$$\begin{aligned} \gamma_0^{(k)}(a) \gamma_2^{(k)}(a) - (\gamma_1^{(k)}(a))^2 + H_k^{(2)}(\gamma_0^{(k)}(a))^2 \\ > (k!)^2 [a(a+1)]^{-k-1} \log^2(1 + 1/a) > 0. \end{aligned} \quad (11.5)$$

Proof. From (11.3), $M_0 = f_0$, $M_1 = f_1 + H_k f_0$, and $M_2 = f_2 + 2H_k f_1 + (H_k^2 + H_k^{(2)})f_0$. Thus $M_0 M_2 - M_1^2 = f_0 f_2 - f_1^2 + H_k^{(2)} f_0^2$. In (11.4), the pair $(j_1, j_2) = (0, 1)$ contributes the right side divided by $(k!)^2$, and all other pairs contribute strictly positive terms. Multiplication by $(k!)^2$ proves the claim. $\square$

The cancellation of H_k at order two has an all-order explanation. Translation of the moment coordinate by $-H_k$ preserves every Hankel determinant: the binomial change of monomials is triangular with diagonal one and acts by congruence on the Gram matrix. The translated generating function, as an identity of germs at $z = 0$ (and for $|z| < 1$), is

$$e^{-H_k z} \zeta(p - z, a) = \exp \left(\sum_{j \geq 2} \frac{H_k^{(j)}}{j} z^j \right) \sum_{n \geq 0} f_n \frac{z^n}{n!}. \quad (11.6)$$

For example, with $h_j^* = H_k^{(j)}$, the strictly positive $d = 3$ determinant is that of the matrix

$$\begin{pmatrix} f_0 & f_1 & f_2 + h_2^* f_0 \\ f_1 & f_2 + h_2^* f_0 & f_3 + 3h_2^* f_1 + 2h_3^* f_0 \\ f_2 + h_2^* f_0 & f_3 + 3h_2^* f_1 + 2h_3^* f_0 & f_4 + 6h_2^* f_2 + 8h_3^* f_1 + (3(h_2^*)^2 + 6h_4^*)f_0 \end{pmatrix}. \quad (11.7)$$

Equation (11.6) is usually more convenient than expanding larger determinants by hand.

11.1 A sharp asymptotic in derivative order

Fix $a > 0$ and $d \geq 1$. Write

$$A_0 = \prod_{j=0}^{d-1} (a + j), \quad V_0 = \prod_{0 \leq i < j < d} \log^2 \left(\frac{a + j}{a + i} \right), \quad T_0(p) = A_0^{-p} V_0, \quad (11.8)$$

$$\rho_1 = \frac{a + d - 1}{a + d}, \quad C_1 = \prod_{j=0}^{d-2} \left[\frac{\log((a + d)/(a + j))}{\log((a + d - 1)/(a + j))} \right]^2, \quad \rho_2 = \frac{a + d - 1}{a + d + 1}. \quad (11.9)$$

All empty products, including those for $d = 1$, equal one.

Theorem 11.3 (Two leading exponential terms and a rigorous remainder). *As $p \rightarrow +\infty$, with a, d fixed,*

$$\frac{D_d(p, a)}{T_0(p)} = 1 + C_1 \rho_1^p + O(\rho_2^p). \quad (11.10)$$

More precisely, for $p \geq p_0 > 1$,

$$0 \leq \frac{D_d(p, a)}{T_0(p)} - 1 - C_1 \rho_1^p \leq E_d(p_0, a) \rho_2^{p-p_0}, \quad (11.11)$$

where the explicit positive constant is

$$E_d(p_0, a) = \frac{D_d(p_0, a)}{T_0(p_0)} - 1 - C_1 \rho_1^{p_0} > 0. \quad (11.12)$$

Proof. The smallest tuple product in (11.4) is uniquely that of $(0, \dots, d-1)$, contributing $T_0(p)$. The next tuple is $(0, \dots, d-2, d)$, whose normalized term is $C_1 \rho_1^p$. Among the remaining tuples, changing only the last coordinate gives the smallest product at $(0, \dots, d-2, d+1)$, with base ρ_2 . For $d \geq 2$,

changing an earlier coordinate forces the last two to be at least $d - 1, d$; its product ratio relative to the first tuple is at least $(a + d)/(a + d - 2)$. This is strictly greater than $(a + d + 1)/(a + d - 1)$, the ratio for the third tuple. The assertion for $d = 1$ is immediate. Hence every omitted normalized base b satisfies $0 < b \leq \rho_2 < 1$.

Write its positive normalized Vandermonde weight as c . For $p \geq p_0$, one has $cb^p \leq cb^{p_0} \rho_2^{p-p_0}$. Sum this inequality over the omitted tuples. Their sum at p_0 is exactly (11.12), proving (11.11) and then (11.10). Strict positivity of E_d follows from the third tuple itself. $\square$

For the Stieltjes derivative tower take $p = k + 1$ and, for example, $p_0 = 2$. This is a rigorous large-derivative-order asymptotic for the normalized nonlinear combinations in (11.3). It is not stated uniformly when a or d grows. Higher exponential terms follow from the exact positive tuple sum; equal products from different later tuples must be grouped if one orders the expansion by distinct bases.

An elementary consequence of the entire generating function, separate from positivity, is

$$\sum_{n \geq 0} \gamma_n^{(k)}(a) \frac{j^n}{n!} = \begin{cases} 0, & j = 1, \dots, k - 1, \\ -(k - 1)!, & j = k. \end{cases} \quad (11.13)$$

For $j < k$, a polynomial zero in (11.2) multiplies a finite Hurwitz value. For $j = k$, cancel the simple pole before taking the limit; the residue gives $-(k - 1)!$. The convergence here is actual Taylor convergence of an entire function.

12 Proposed corrections to the current manuscript

The two proposed edits below are limited to `Analysis/Polylogarithms/docs/manuscript/chapters/07-integration.tex` at the pinned snapshot. The package includes a minimal unified patch, with context and source hash, so the changes can be reviewed separately from adding the present article. Neither edit changes the coefficients of an existing special-function identity.

12.1 The Hurwitz ladder needs an explicit convergence domain

At the label `integral:res:hurwitzladder`, the manuscript prints

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + \frac{1}{2} \sum_{n \geq 2} \frac{n - 1}{n(n + 1)} \zeta(n, z + 1). \quad (12.1)$$

The formula is correct on $\operatorname{Re} z > 0$, with the analytic branch of $\log \Gamma$ normalized on the positive real axis and the principal $\log z$. That sufficient domain should be stated. The unqualified series is not valid at all regular gamma arguments: at $z = -1/2$, $\zeta(n, 1/2) = (2^n - 1)\zeta(n)$, so its summands do not even tend to zero. This is a convergence-domain omission, not a wrong-coefficient claim.

Proposition 12.1 (Convergence and an explicit ladder remainder). *Equation (12.1) holds with locally uniform absolute convergence on $\operatorname{Re} z > 0$. If $N \geq 2$ and $\operatorname{Re} z \geq \sigma > 0$, the tail after $n = N$ satisfies*

$$|\operatorname{tail}_N(z)| \leq \frac{\zeta(N, \sigma + 1)}{2(N + 1)\sigma} \leq \frac{(\sigma + 1)^{-N} + (\sigma + 1)^{1-N}/(N - 1)}{2(N + 1)\sigma}. \quad (12.2)$$

Proof. For real $w > 1$, elementary summation of the power series gives

$$\frac{1}{2} \sum_{n \geq 2} \frac{n - 1}{n(n + 1)} w^{-n} = (w - \frac{1}{2}) \log \frac{w}{w - 1} - 1. \quad (12.3)$$

For instance, use $(n-1)/(n(n+1)) = -1/n + 2/(n+1)$ and the logarithmic power series. For real $z > 0$, the positive terms permit interchange with $\zeta(n, z+1) = \sum_{m \geq 0} (z+m+1)^{-n}$. Let

$$B(z) = \log \Gamma(z) - (z - \tfrac{1}{2}) \log z + z - \tfrac{1}{2} \log(2\pi).$$

The gamma recurrence gives $B(z) - B(z+1) = (z+1/2) \log((z+1)/z) - 1$. Thus the sum of (12.3) over $w = z+m+1$, $0 \leq m < M$, equals $B(z) - B(z+M)$. Stirling's formula implies $B(z+M) \rightarrow 0$, proving (12.1) on the positive axis.

For $\operatorname{Re} z \geq \sigma > 0$ and $n > N$, $(n-1)/(n(n+1)) \leq 1/(N+1)$ and $|z+m+1|^{-n} \leq (\sigma+m+1)^{-n}$. Summing the resulting geometric series gives

$$|\operatorname{tail}_N(z)| \leq \frac{1}{2(N+1)} \sum_{m \geq 0} \frac{(\sigma+m+1)^{-N}}{\sigma+m} \leq \frac{\zeta(N, \sigma+1)}{2(N+1)\sigma}.$$

The elementary integral bound for a decreasing power series yields the second inequality in (12.2). These estimates prove local uniform absolute convergence, so the identity theorem extends the positive-axis formula to the right half-plane. $\square$

The proposed source edit inserts the domain and branch convention before the displayed ladder and a brief convergence statement afterward. The detailed proof and reusable error bound remain in this article.

12.2 Integer-relation searches do not require an independence theorem

At `integral:neg:psim2`, the sentence requiring the basis atoms to be $\mathbb{Q}$ -independent is too strong. The observed issue is that known dependencies among the background coordinates can make a one-relation search return a relation whose target coefficient is zero. Removing known dependencies is good practice; proving actual arithmetic independence of the remaining constants is not a prerequisite for a valid search.

The proposed replacement is:

Remove known dependencies from the background coordinates, or retain them explicitly and search modulo their known relation space. A returned relation must have a nonzero target coefficient to evaluate the target. No unproved arithmetic-independence assumption is needed.

This is a correction to the methodological wording. It does not change the recorded failed experiment or rehabilitate its rejected candidate. The current discovery chapter already distinguishes formal coordinates from arithmetic independence, so the edit also makes the explanations consistent across the manuscript.

13 Further research questions

The following questions separate tractable extensions from statements that need substantially new arguments. They are proposals for further work; none is used as an assumption in the proofs above.

- 1. Products at three or more independent shifts.** Determine an explicit generating function for

$$\operatorname{FP} \int_0^1 \prod_{j=1}^r \gamma_{n_j}(\{x + a_j\}) \, dx, \quad r \geq 3,$$

first with distinct shifts and then as selected shifts coalesce. These are simultaneous point-wise products, not iterated convolutions, which are already controlled by Theorem 6.1. The Fourier constraint now leaves several summation indices, suggesting multiple polylogarithms and Tornheim-type derivatives. A useful first goal is a complete three-factor subtraction formula with every local counterterm specified.

2. **Uniform collision asymptotics with growing jet order.** Theorem 10.1 fixes p, r , and the finite-part theorem fixes n, m . Obtain estimates uniform when an index grows while $a \rightarrow 0$. The competition between factorial coefficient growth, gamma singularities in the generating variables, and powers of $\log a$ is absent from the fixed-index arguments. An explicit transition scaling and error bound would make the identities useful at high weight.
3. **Sign and extremum classification for higher jet correlations.** The full positivity of the even derivatives of R is stronger than positive definiteness of an autocorrelation kernel. Decide which $R_{p,r}$, or which derivatives and balanced combinations of them, retain an analogous sign pattern. For $p > 0$, the Bernoulli cases already have a different pattern, so the log-Gamma conclusion should not be extrapolated without proof. The explicit coefficients in (10.4) give a precise starting point for counterexamples or classification theorems.
4. **Optimal inequalities for higher Stieltjes reflection sums.** The sharp midpoint inequality (9.12) treats index one. For $n \geq 2$, determine the extrema and sign changes of $\gamma_n(a) + \gamma_n(1-a)$ and of the universal lower-index corrections appearing in $F_{n,m}^+$. Formulate the problem at the level of complete correlation derivatives so that any endpoint singular terms are retained.
5. **Rational shifts, characters, and exact relation spaces.** Combine Theorem 5.2 with the manuscript's character and conductor descent. At a fixed conductor and total index, compute the formal span of the resulting bilinear identities and its exact relations. Separate this finite symbolic quotient from the much harder question of arithmetic independence of the evaluated constants. Useful deliverables would be small, replayable rational certificates, not only reduced decimal expressions.
6. **Normalization under rescaling and nonlinear changes of variable.** Theorem 7.1 fixes one coordinate and scale. Classify how the finite parts and their Dirac derivatives transform when the period or local coordinate changes. Determine which combinations of zero mean, reflection symmetry, differentiation, and the Hurwitz semigroup can be preserved simultaneously. This would give a principled way to compare different regularization conventions used in symbolic integrations.
7. **Large Hankel dimension and joint parameter limits.** Theorem 11.3 fixes a, d . Find a uniform asymptotic when d grows with p , or when a also varies. The exact positive Vandermonde sum provides explicit candidates for the dominant tuples, but control of their number and near-equal products is required. Bounds for the smallest eigenvalue would turn determinant positivity into quantitative stability estimates for the derivative tower.
8. **Certified high-order evaluation.** Turn the coefficient algorithms and Cauchy order-differentiation checks into interval enclosures. For a finite-radius contour, derive an explicit trapezoidal aliasing bound from a larger analytic annulus and include the evaluation errors at every node. Combine it with (9.7) and the finite-part endpoint subtractions. This would give certified numerical values without mistaking working precision for an error certificate.

- 9. The remaining mixed Gaussian candidates.** The frozen S_6 relation and the new S_8 vector in `s8new:conj:S8` remain genuine targets. Seek an exact convergent word or shuffle certificate, building on the already proved S_4 transformation, or a rigorous nonzero residual. An interval containing zero, however narrow, proves proximity only. The earlier rejected S_8 vector must remain distinct from the new candidate. The present convolution identities supply additional analytic representations but do not provide those missing certificates.
- 10. A precise reduction question for higher log-Gamma moments.** The manuscript already has a Tornheim representation for the cubic moment. Fix an explicit candidate algebra of single zeta and Dirichlet L derivatives and ask whether a reduction into it can be proved. Shifted three-point correlations may expose useful functional equations before taking the coincident-shift limit. Failure of a chosen finite relation search would not establish irreducibility in that algebra.

14 Verification, reproducibility, and integration status

The proofs establish the identities. Computations serve three narrower purposes: they check finite symbolic expansions, compare independent analytic representations numerically, and make the proposed source edits reproducible. They are not used to infer an infinite identity from a small residual.

14.1 Exact algebra and independent numerical checks

The file `verification/exact_coefficients.py` expands the gamma germs in a rational polynomial ring with formal symbols for zeta values and Stieltjes functions. It checks 60 exact assertions and exports 15 coefficient pairs, covering all $n + m \leq 4$ in both orientations. No floating-point arithmetic or integer-relation search enters those calculations. The companion `verification/exact_jet_polynomials.py` checks the displayed jet collision polynomials and their logarithmic derivatives symbolically. These finite checks complement, rather than replace, the every-index analytic proofs.

The numerical scripts use independent left and right sides where practical. In particular, the finite-part integrals use ordinary real quadrature after explicit local subtraction. The following values are representative; all digits and settings are recorded in the JSON files.

Quantity	Value, rounded	Typical absolute discrepancy
$F_{0,0}^+(1/2)$	-5.99678749530633590836	below 10^{-53}
$F_{0,0}^-(1/2)$	-1.06198529476165659894	below 10^{-53}
$F_{1,0}^+(\sqrt{2}-1)$	6.90546420565063770565	below 10^{-41}
$F_{1,0}^-(\sqrt{2}-1)$	-2.43190584182042316367	below 10^{-41}
$R(0)$	1.86631708379356208099	below 10^{-45}
$R(1/2)$	0.52595583980330838858	below 10^{-45}

The finite-part files contain eleven zeroth-order checks at 55-digit working precision and four mixed-index checks at 45 digits. The ordinary correlation file contains sixteen checks at 48 digits, including real parameters with $\operatorname{Re}(s+t) > 1$, a complex critical-order example, and direct log-Gamma quadrature. This specifically tests the separated singularity domain in Theorem 3.1.

The raw derivative finite-part check uses (7.7) at $a = 1/2$ and $a = 0.37$, with cutoffs down to 10^{-18} and 65 working digits. Its unextrapolated discrepancy is of order ϵ , as expected. A Taylor expansion of the two omitted endpoint intervals gives the first error coefficient

$$C(a) = \gamma\psi'(a) - \frac{3}{2}\psi''(a) - \zeta(2)\psi(a).$$

Subtracting $C(a)\epsilon$ leaves discrepancies below $1.2 \cdot 10^{-34}$ in the two final rows. These smaller errors reflect explicit cutoff acceleration, not an interval proof of the integral value.

The Hankel/ladder file uses 80-digit arithmetic. It checks compensated Stieltjes derivatives through index four for three parameter choices, strict $d = 2, 3$ determinant bounds, finite Cauchy–Binet sums, twenty tuple-order cases, thirty asymptotic remainder bounds, and six real or complex Hurwitz-ladder examples. All use the domains and normalizations stated in the theorems.

14.2 A numerical implementation issue found during verification

In the installed `mpmath` 1.3.0, applying the default infinitesimal `mp.diff` directly to a boundary polylogarithm at an integer order can lose accuracy in a precision-dependent way. One log-Gamma check at 48 working digits had a discrepancy about $4.5 \cdot 10^{-13}$, while finite-radius Cauchy differentiation agreed with independent real quadrature to about 10^{-49} . A similar problem occurred in an order-zero second derivative at a different working precision. These observations concern that evaluation path; they do not refute the polylogarithm identities.

The final checks therefore use the Cauchy coefficient formula

$$f^{(r)}(w_0) \approx \frac{r!}{N\rho^r} \sum_{j=0}^{N-1} f(w_0 + \rho e^{i\theta_j}) e^{-ir\theta_j}, \quad \theta_j = \frac{2\pi(j+1/2)}{N}, \quad (14.1)$$

with $\rho = 1/8$ and $N = 64$, instead of the vulnerable path. The full finite-contour remainder is not interval-certified in these scripts. Its agreement with separately subtracted real quadrature is numerical evidence, clearly labeled as such in the outputs. The direct `mp.diff` comparison is retained only as a diagnostic of the implementation issue, not as a theorem-check acceptance test.

14.3 Delivered sources and proposed integration

The archive contains the complete standalone T_EX source, compiled PDF, the correlation figure in PDF and PNG, runnable Python scripts, JSON results, a build recipe, source provenance, and the minimal correction patch. The source manifest records the pinned Git commit and hashes of the repository files consulted. No repository change is implied by delivery of the patch.

For integration, the most natural route is a bilinear-calculus section after the existing Stieltjes derivative and antiderivative material, with a cross-reference from the log-Gamma moment chapter. The Hankel section can also stand as a separate derivative-tower continuation. The standalone labels should receive a common prefix when merged with the main manuscript; the accompanying integration note gives the specific dependency order. Classical formulas should retain their attribution, the finite-part convention must accompany the displayed integrals, and the distributional contact terms must accompany global convolution identities.

A Unequal jets and resonant collision cancellation

For completeness, the shifted master kernel gives an explicit route to unequal spectral orders as well. Put $\sigma = s + t$ and suppose initially $\operatorname{Re} s, \operatorname{Re} t < 1$ and $\sigma \notin \{1, 0, -1, \dots\}$. The unsymmetrized local polylogarithm expansion gives the convergent identity, for $0 < a < 1$,

$$\begin{aligned} \mathcal{K}_+(s, t; a) &= \frac{\Gamma(1-s)\Gamma(s+t-1)}{\Gamma(t)} a^{1-s-t} \\ &\quad + 2\Gamma(1-s)\Gamma(1-t)(2\pi)^{s+t-2} \sum_{j \geq 0} \frac{\zeta(2-s-t-j)}{j!} \cos \frac{\pi(s-t-j)}{2} (2\pi a)^j. \end{aligned} \quad (\text{A.1})$$

The coefficient of the fractional power is $B(1-s, s+t-1)$ by analytic continuation. It also follows independently from the local scaling integral of $x^{-s}(x+a)^{-t}$ where that integral converges.

At $\sigma = 1 - M$ with $M \geq 0$, the fractional term and the $j = M$ term in the sum can have poles separately. They must be combined before taking the limit or differentiating. Here is an explicit holomorphic prescription for all unequal negative-integer jets. For integers $p, q \geq 0$, put $s = -p + u$, $t = -q + v$, $M = p + q + 1$, $w = u + v$, and let

$$b(u, v) = w B(1-s, s+t-1),$$

while $d(u, v)$ is w times the coefficient of a^M in the analytic series in (A.1). Both are holomorphic near zero. Since the full kernel is holomorphic, their residues cancel: $b(u, -u) + d(u, -u) = 0$. The resonant pair equals

$$a^M \left[b(u, v) \frac{e^{-w \log a} - 1}{w} + \frac{b(u, v) + d(u, v)}{w} \right]. \quad (\text{A.2})$$

Both quotients are holomorphic. Taylor extraction now gives unequal jet orders with no differentiation of an uncanceled pole. The symmetric equal-order theorem in Section 10 is the particularly simple case where the cancellation is already absorbed into the beta/cosine germ.

If also $\text{Re}(s+t) < 1$, the singularities may coincide in the ordinary integral and

$$\begin{aligned} \mathcal{K}_+(s, t; 0) &= \int_0^1 \zeta(s, x) \zeta(t, x) dx \\ &= 2\Gamma(1-s)\Gamma(1-t)(2\pi)^{s+t-2} \cos \frac{\pi(s-t)}{2} \zeta(2-s-t). \end{aligned} \quad (\text{A.3})$$

This additional condition controls the generic product singularity x^{-s-t} ; it is not needed at a fixed nonzero shift. At the other seam, $a \rightarrow 1$, interchange s and t . These distinctions explain why an unshifted product formula cannot simply be used to infer a shifted integrability restriction.

References

- [1] V. Reshetnikov, *ProveIt*, polylogarithm manuscript, editorial ledger, and incoming archive register, snapshot 570b0567f311cf1890865065896be2665f469e4f, 10 October 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/570b0567f311cf1890865065896be2665f469e4f/Analysis/Polylogarithms/docs/manuscript>. Incoming register: <https://github.com/VladimirReshetnikov/ProveIt/tree/570b0567f311cf1890865065896be2665f469e4f/docs/incoming>.
- [2] NIST Digital Library of Mathematical Functions, §25.11, *Hurwitz Zeta Function*; in particular 25.11.3, 25.11.9, 25.11.12, and the parameter and spectral derivative formulas. <https://dlmf.nist.gov/25.11>. Accessed 10 October 2026.
- [3] NIST Digital Library of Mathematical Functions, §25.12, *Polylogarithms*; especially the local expansion 25.12.12. <https://dlmf.nist.gov/25.12>. Accessed 10 October 2026.
- [4] NIST Digital Library of Mathematical Functions, §25.4, *Reflection Formulas*; especially 25.4.2. <https://dlmf.nist.gov/25.4>. Accessed 10 October 2026.
- [5] O. Espinosa and V. H. Moll, On some integrals involving the Hurwitz zeta function: Part 1, *The Ramanujan Journal* **6** (2002), 159–188. https://www.math.tulane.edu/~vhm/papers_html/hurwitz1.pdf.

- [6] Z.-H. Sun, On the properties of invariant functions, *Bulletin des Sciences Mathématiques* **189** (2023), 103347. Primary preprint arXiv:2209.14625v1 (2022), Remark 3.2, printed p. 15. <https://arxiv.org/pdf/2209.14625v1>.
- [7] H. Zhu, S. Hu, and M.-S. Kim, On the properties of alternating invariant functions, arXiv:2505.10985v1 (2025), Theorem 1.6. <https://arxiv.org/html/2505.10985v1>.
- [8] R. Gay and A. Sebbar, Pseudo-differential operators on the circle, Bernoulli polynomials, *Quantum Studies: Mathematics and Foundations* **11** (2024), 1–25. <https://doi.org/10.1007/s40509-024-00316-9>.
- [9] K. Dilcher, On generalized gamma functions related to the Laurent coefficients of the Riemann zeta function, *Aequationes Mathematicae* **48** (1994), 55–85. <https://doi.org/10.1007/BF01837979>. The bibliographic record was consulted; the exact series attribution used here is verified through [10, printed p. 16].
- [10] M. Brede, *Eine Reihenentwicklung der vervollständigten und ergänzten Riemannschen Zeta-funktion und Verwandtes*, doctoral dissertation, Universität Gesamthochschule Kassel, 2001; printed p. 16 for the series attributed to Dilcher. <https://www.mathematik.uni-kassel.de/~koepf/Diplome/brede.pdf>.